



AMSC5720
Advanced Materials Synthesis
(先進材料合成)

Course Information

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Lecture Time and Venue

- ◆ **Lectures**

- ◆ Every Tuesday 18:30-21:15

- ◆ **Venue**

- ◆ Wu Ho Man Yuen Building (WMY) 507 (5/F) (WMY_507)

- ◆ Language: English

- ◆ Units: 3



Topics

- ◆ ***Topic 0: Basic Concepts of Structure and Properties of Materials***
- ◆ **Topic 1:** Fabrication of Classic Materials (Metals, Ceramics, Polymers, Composites)
- ◆ **Topic 2:** Advanced Materials Synthesis (Semiconductors, Nanomaterials, Smart Materials, Additive Manufacturing, Biomaterials, Battery Materials, etc.)
- ◆ **Topic 3:** Representative Advanced Materials & Case Studies
- ◆ **Topic 4:** Introduction to Quality Control



Assessment

- ◆ **Assignments (20%, 10% each)**
 - ◆ Assignment 1
 - ◆ Assignment 2
- ◆ **Project (35%)**
- ◆ **Final Exam (45%)**
 - ◆ Centralized (RES will inform date/venue), cover all topics



Detailed Schedule (tentative)

Week No.	Tuesday, 18:30-21:15	Activity	Topic
1	Sept 8	Introduction, Lecture	0
2	Sept 15	Lecture	0
3	Sept 22	Lecture	1
4	Sept 29	Lecture	1
5	Oct 6	Lecture	1
6	Oct 13	Lecture	2
7	Oct 20	Lecture	2
8	Oct 27	Lecture	2
9	Nov 3	Lecture/Guest Lecture	3
10	Nov 10	Lecture/Guest Lecture	3
11	Nov 17	Lecture	4
12	Nov 24	Project Assessment	-
13	Dec 1	Project Assessment/Revision	-



TA Contacts for Prof. Chen's Part

- ◆ Mr. XIAO Zetao (1155248135@link.cuhk.edu.hk)
- ◆ Ms. GENG Nannan (1155277454@link.cuhk.edu.hk)
- ◆ Mr. CHEN Zhengjie (1155282884@link.cuhk.edu.hk)



Project Assessment

- ◆ A mini literature study of advanced materials fabrication. A topic list of various different kinds of frontier materials will be given for selection.
- ◆ **Assessment method:** a(n) individual/group presentation with QnA to be given to the class.
 - ◆ Details to be announced
- ◆ **Mode I:** Individual Project
- ◆ **Mode II:** Group Project + Group/Individual Marking

